Commercial Card Games Improve Middle School Students' Executive Functioning

Department of EDUCATIONAL PSYCHOLOGY

Cristaly Mercado, McNair Scholar • Keisha Varma, Ph.D.
Jesslyn Valerie • Jimin Park • Christina Zdawczyk • Sashank Varma, Ph.D.

CEHD College of Education + Human Development
University of Minnesota
Driven to Discover℠

Introduction

Improving your score at a card game to defeat your friends requires practice. You not only practice the game's strategies, but also improve the mental processes needed for gameplay. Games and executive functioning can both be practiced and improved upon: perhaps simultaneously.

Background

Executive Functioning (EF) refers to the cognitive abilities that allows people to adapt in order to meet their goals.

EF hosts the following skills (Miyake et. al, 2000):

shifting	switching between mental processes
updating	monitoring and updating information in working memory
inhibition	suppressing typical responses

EF is a significant predictor for Middle School academic achievement (Samuels, et al., 2016).

Shifting ability has been shown to be related to science achievement and to science learning in middle and high school students (K. Varma, Van Boekel, & S. Varma, 2018).

A student who struggles with inhibition -or any other EF skill- does not need to do so for their entire life. EF can be improved through intervention and training (Diamond & Lee, 2011).

Research Questions

1. Do students show increased performance on measures of executive function after playing games

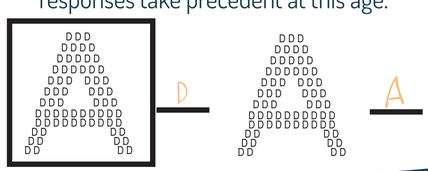
2. Is there a relationship between students' performance on specific scientific practices targeted in the games and measures of EF?

Measures

A flanker task and anti-saccade task was used to assess inhibition.



A local global task assessed shifting ability, as global responses take precedent at this age.

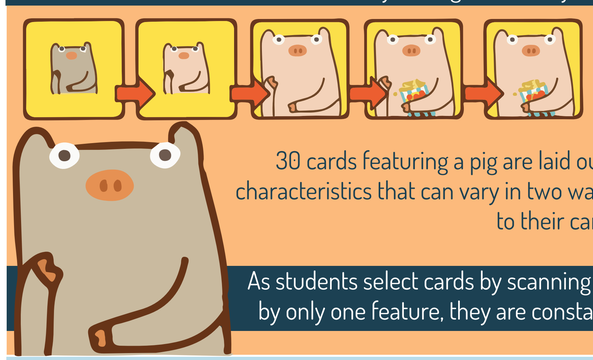


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Methods

Participants: This study took place in a large, urban middle school located in the Upper Midwest. The seventh graders who participated were relatively high-achieving students.

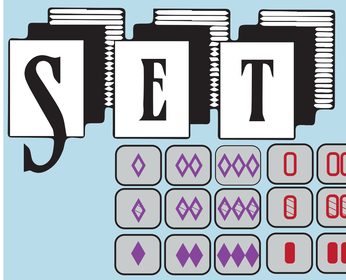
Single Group Pre-Post-Test Research Design: Students were given Executive Functioning pre-tests. Students were randomly divided into groups of 4 to play four games with each other for 3 class periods. They were given a delayed post-test two weeks later.



Pick-a-PIG

30 cards featuring a pig are laid out in a grid. Each player starts with one card. Pigs have 5 characteristics that can vary in two ways. Players select a card from the grid that is either identical to their card or differs by only one feature.

As students select cards by scanning and determining whether they match or differ by only one feature, they are constantly shifting their attention between features.



SET

Each card has one to three of the same object. Objects can vary in three different ways: shape, shading and color. Players simultaneously attempt to collect sets of cards from 12 that are laid out. A 'set' is three cards that must either differ or all be the same for each feature across cards. The player that has collected the most sets at the end of the game wins.

Example of a Set:

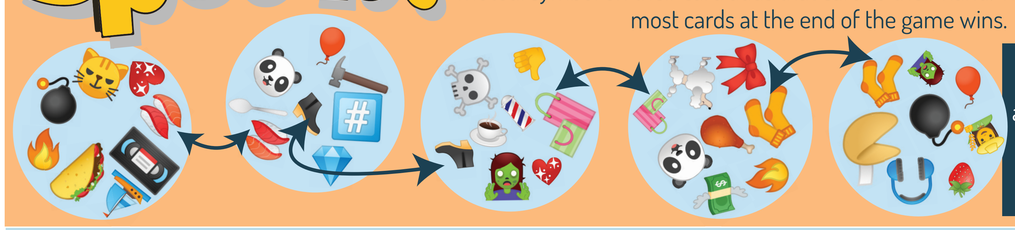


Students must shift their attention between the four categories of features and focus their attention on the features that will form their sets of cards.

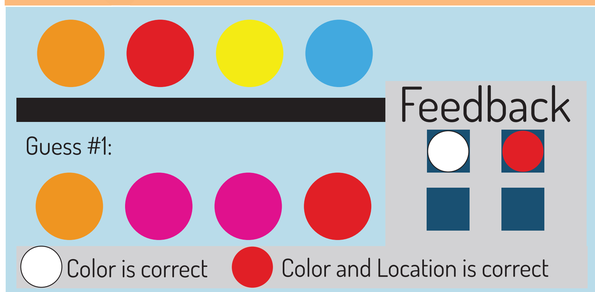


Spot it!

Each card features 8 different symbols that vary in size and location. Any two cards will have one symbol in common. All the players receive a card, a card is laid out in the center of the table. The first player to identify and name the mutual symbol on their card adds that card to their hand. The player with the most cards at the end of the game wins.



Students must focus their attention on each symbol on the cards to find the common one.



Players must shift their attention between the guess and the feedback as the game progresses.

MASTERMIND

Students play in pairs. The codemaker places a series of 4 pegs in the secret code area of the gameboard. The Codebreaker tries to guess what the code is by placing pegs in the first row. The Codemaker tells the codebreaker how many are the right color and how many are in the right position.

Results

Flanker Anti-Saccade		Pre-Test	Post-Test
n=47 m ± SD	Time (in seconds)	122.34 ± 27.06	97.77 ±19.95
	Correct Responses	95.21 ±1.02	94.02 ± 9.18

On the Flanker and antisaccade tasks, students reported significantly faster completion times on the posttests. However, they did not improve in the number of items they answered correctly.

Local Global		Pre-Test	Post-Test
n=47 m ± SD	Correct Responses	30.2 ± 9	36 ± 9.7
	Incorrect Responses	1.9 ± 2.8	.34 ± 0.7

On the shifting measure, students performed significantly better on the local-global task after the gameplay experiences. They answered significantly more items correctly. They also made fewer errors, however, that difference is only marginal.

Conclusion + Discussion

Students improved their executive functioning after playing games with their classmates for several days. Their desire to win the game naturally resulted in EF training without the need for heavy-handed intervention.

Thus, EF improvement is not exclusive to researcher-designed training. The commercial games the students played are marketed as entertainment games and produced enmasse. This can make EF training more accessible to families and schools.

Future studies should assess the potential effects of games that students are already playing in their free time have on EF.

References

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